Jake Doherty

Austin, TX

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Profile

Ph.D. candidate in Physics with expertise in numerical optimization, scientific computing, algorithm development, and large-scale data analysis. Extensive experience designing high-performance C++ and Python software for solving nonlinear mathematical models and analyzing large computational datasets.

Education

University of Texas at Austin 2022–Present

PhD Candidate, Physics

University of South Florida May 2022

B.S. Physics, B.A. Mathematics

Technical Skills

Programming: Python, C++, C, Fortran, Bash

Scientific Computing: NumPy, SciPy, Matplotlib, Linux/Unix, Slurm, Git

Methods: Numerical integration of differential equations, Numerical methods, Data analysis, Mathematical modeling

High Performance Computing: Parallel and cluster computing, Large-scale numerical simulations

Research Experience

Graduate Researcher, University of Texas at Austin Aug 2022–Present

Advisor: Pablo Laguna

- Design and implement numerical algorithms for solving coupled nonlinear partial differential equations using C++ and Fortran.
- Run large-scale simulations on HPC clusters using distributed computing resources.
- Analyze very large simulation datasets and build automated post-processing pipelines using Bash scripts and Python.

Research Assistant, University of South Florida May 2020–Aug 2022

Advisor: Inna Ponomareva

- Modeled ferroelectric and antiferroelectric materials using computational physics methods.
- Performed numerical analysis of polarization, phase transitions, and electronic structure.

Selected Projects

Head-on Collisions of Boson Stars with Bowen–York Type Initial Data

- Developed a mathematical algorithm to construct constraint satisfying initial data for subsequent numerical evolution, and implemented said algorithm via modification of several C++ libraries.
- Implemented shooting method for solving coupled nonlinear PDEs
- Constructed piecewise cubic Hermite interpolating polynomials to interpolate among solutions.

Black Hole Growth in Scalar Field Cosmology

- Performed large scale numerical integration of PDEs on HPC clusters to study black hole spacetimes.
- Implemented numerical algorithms for manipulation of large datasets, including interpolation and plotting.

Publications

arXiv:2607.09494 [gr-qc] (2026)
Head-on Collisions of Boson Stars with Bowen–York Type Initial Data **(First Author)**

Physical Review D 112, 044029 (2025)
Growth of a Black Hole in a Scalar Field Cosmology **(First Author)**

Journal of Physical Chemistry C 125(29), 16296–16303 (2021)
Tunability of Structure, Polarization, and Band Gap of High-T_c Organic-Inorganic Ferroelectrics by Hydrostatic Pressure

Journal of Applied Physics 132, 034101 (2022)
Negative Capacitance Regime in Antiferroelectric $PbZrO_3$ **(First Author)**

Talks/Conferences

10th Gulf Coast Gravity Meeting	03/2024
APS Global Physics Summit	03/2026

Grants/Awards

Texas Advanced Computing Center Allocation – Compact Object Mergers and Gravitational Waves	2026
USF Dean's List of Scholars	2021
USF Director's Award	2018
Florida Academic Scholars Award	2018

Leadership & Teaching

Teaching Assistant, University of Texas at Austin	2022–Present
Directed Reading Program Mentor, University of Texas at Austin	2022–Present
Teaching Assistant, University of South Florida	2021–2022